ME/AER 676 Robot Modeling & Control
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Models of Robotic Manipulators

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Tasks For Robots

- What kinds of tasks requires a physical robot?
  (Or, what can't your phone do?)
- How would you describe the motion you want to a robot?
- How would you, as a robot, convert that description into action?

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  A: Joint Coordinates and Joint Torques
  We start with joint coordinates

Robot Configurations: Joint Variables

The configuration q of such robots will depend on the joints between rigid parts that make up the robot

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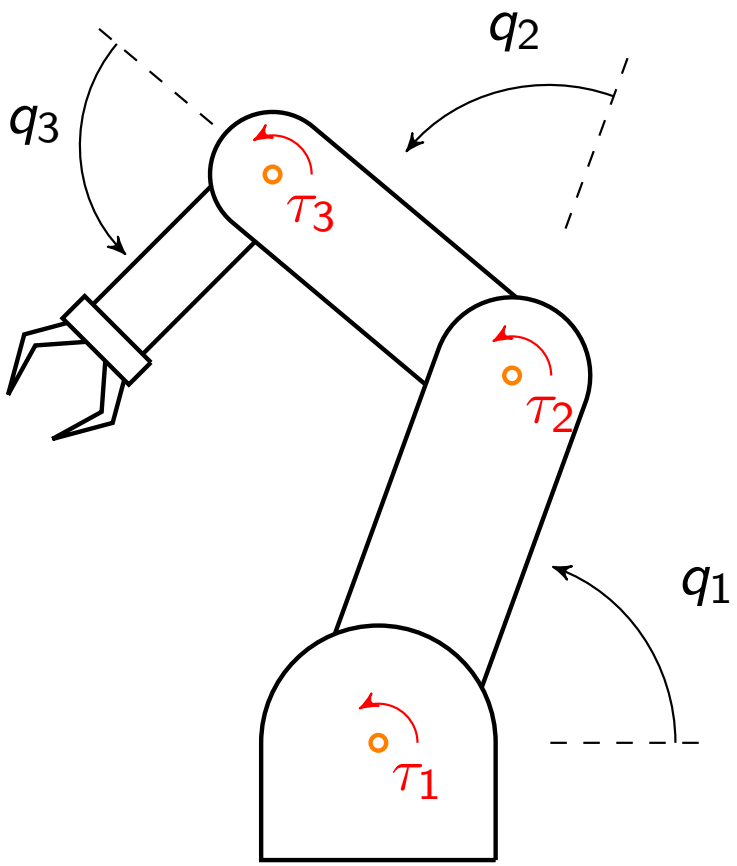

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Dynamics

The configuration q changes with time according to the second-order nonlinear ordinary differential equation

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + \tau_{friction} + \tau_e, \tag{1}$$

where

- $D(q)$: inertia matrix (some formulations use $M(q)$)
- $C(q, \dot{q})$: Coriolis terms
- $G(q)$: the conservative forces (gradient of potential energy)
- τ is the force or torque vector generated by actuators at these joints
- $\tau_{friction}$ represents dissipative forces
- τ_e represents externally applied forces through contact with the environment

State Space Models

How do these dynamics compare to the usual state-space system?

$$\dot{x} = f(x, u)$$
$$y = h(x, u)$$

The configuration $q \in \mathbb{R}^n$ defines a state

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}.$$

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The output y above corresponds to the task space coordinate of the robot.
WARNING: Roboticists do not always use symbols x and y as above.

Goals

For tpython -m tkinterhis section of the course, we will

- Understand joint coordinates q
- Understand properties of the dynamics model (1)
- Understand the relationship $\chi = FK(q)$, where χ is the task coordinates (output) and FK is the forward kinematics map
- Derive controllers that make $q(t) \to q_d(t)$
- Derive controllers that make $\chi(t) \to \chi_d(t)$
- Derive controllers that enforce relationships between external forces τ_e and joint velocity $\dot{q}$ (Energy-based control)
- Convert this learning into simulations
- Understand challenges of perceiving the world

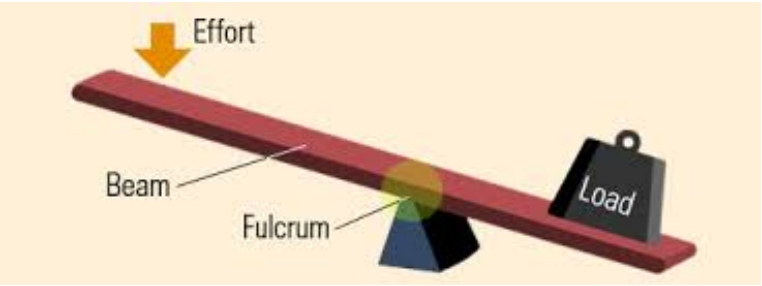
Goals

At the end of this part of the course, you should

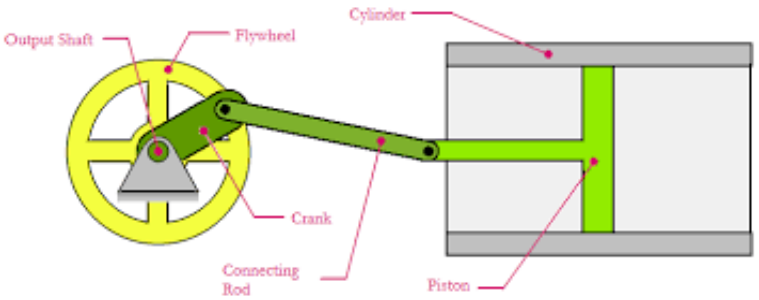
- understand how to derive equations and controllers in principle;
- know how to leverage software libraries to perform calculations involving robot models and controllers.

Robot Mechanisms

Mechanisms such as simple levers and crank-and-piston are widely used to transfer power/energy/force.



Simple lever



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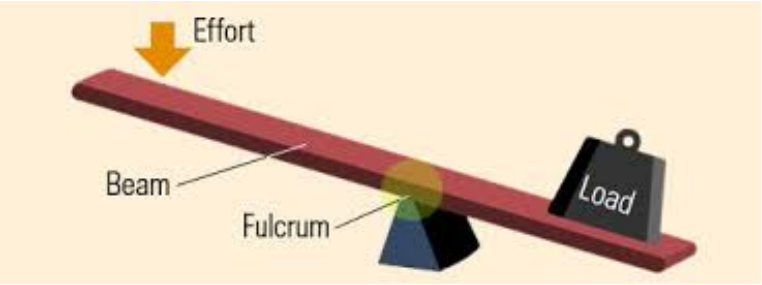

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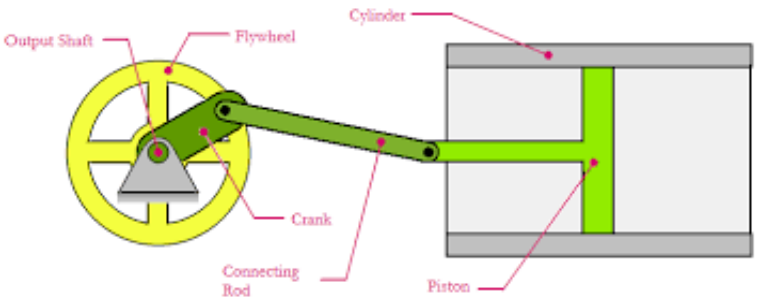


Crank and Piston

These mechanisms consist of (ideally) rigid bodies connected by joints that constrain relative motion.

Examples of Joints

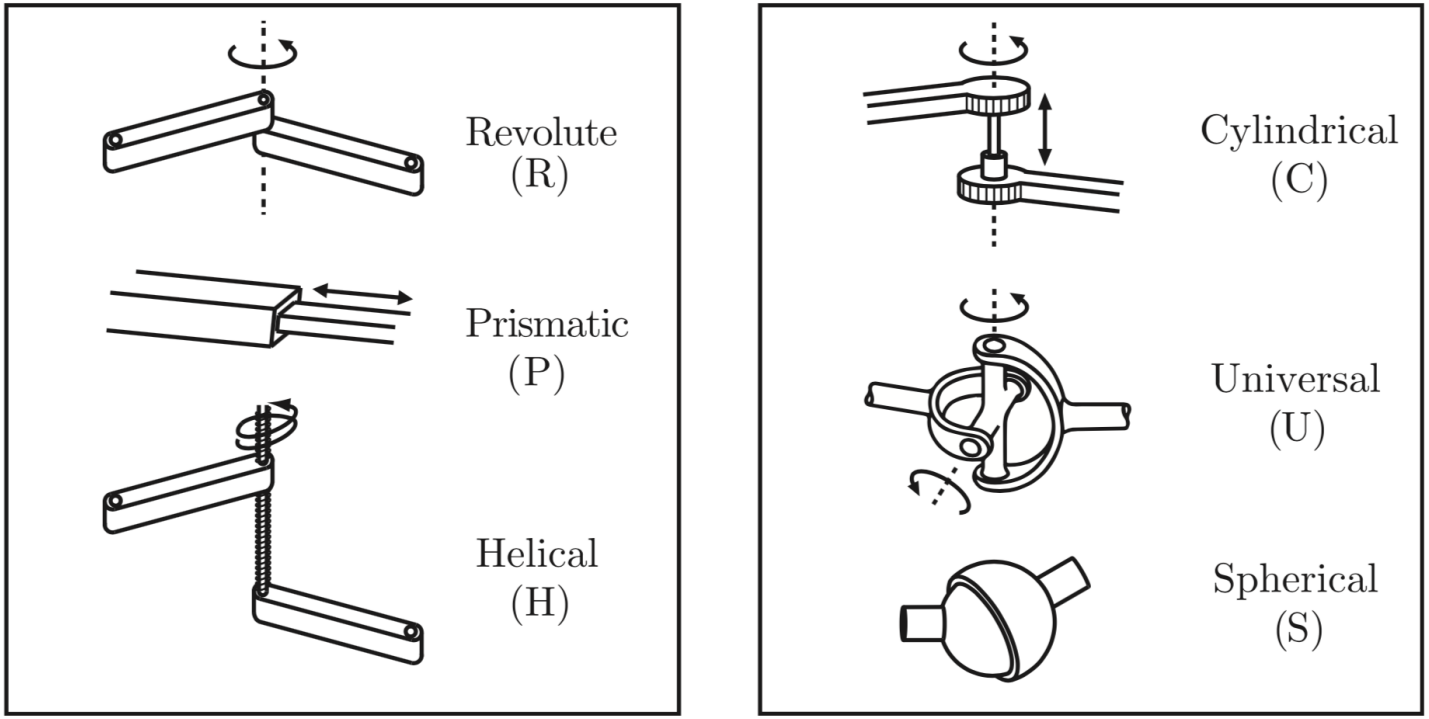
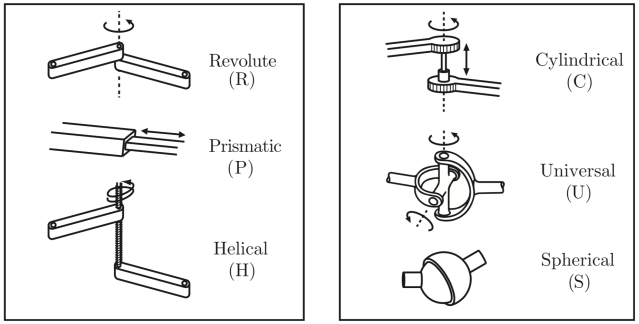


Figure 2.3: Typical robot joints.
source: Modern Robotics

Any joint can be abstractly replaced by multiple revolute (R) and/or prismatic (P) joints. Example: C= R+P

Joint Constraints

Joint type	dof f	Constraints c between two planar rigid bodies	Constraints c between two spatial rigid bodies
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	N/A	5
Cylindrical (C)	2	N/A	4
Universal (U)	2	N/A	4
Spherical (S)	3	N/A	3



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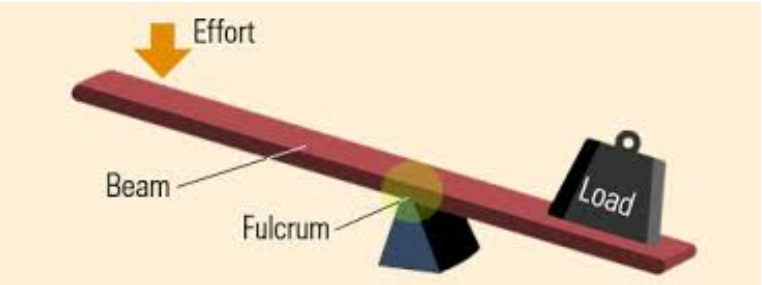
Exercise: Planar mechanisms

How many joints can you identify here? :



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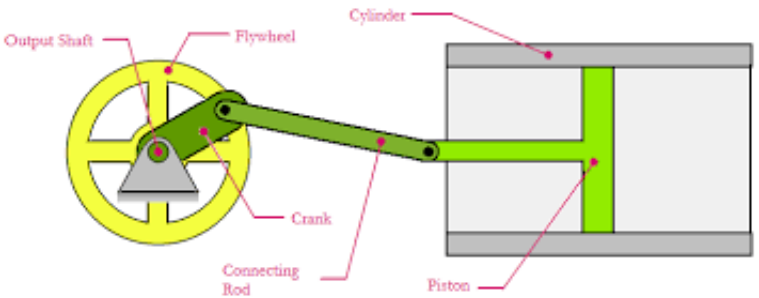

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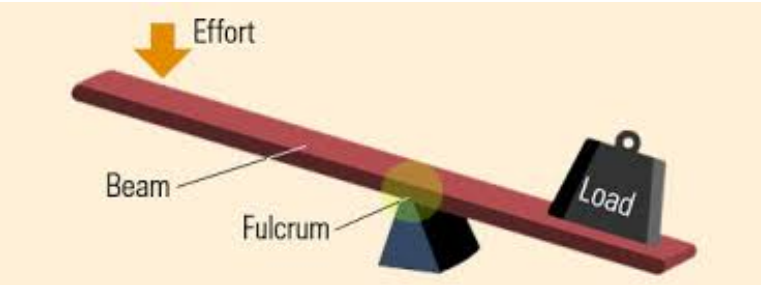

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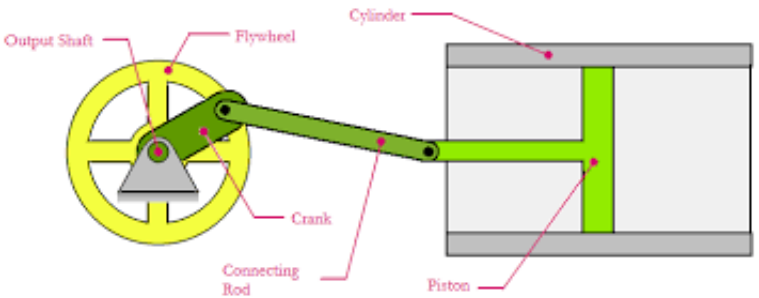
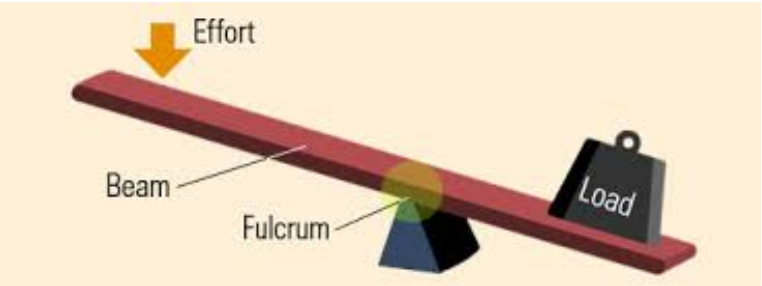


Crank and Piston

Lever: one revolute joint at the fulcrum.
Crank and Piston: Three revolute joints, one prismatic joint (formed by piston and cylinder)

Exercise: Planar mechanisms

What are the degrees of freedom of these mechanisms? :



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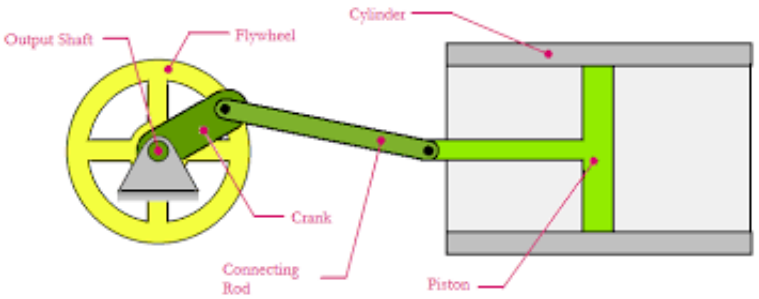

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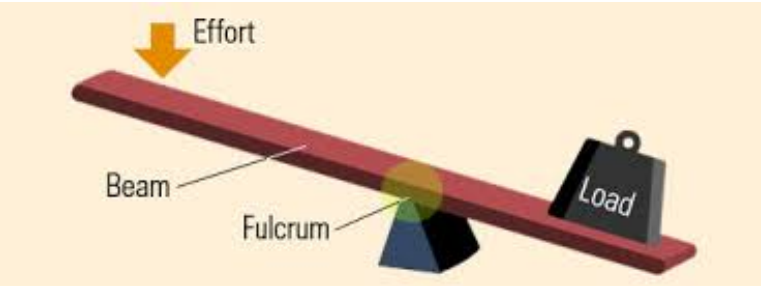

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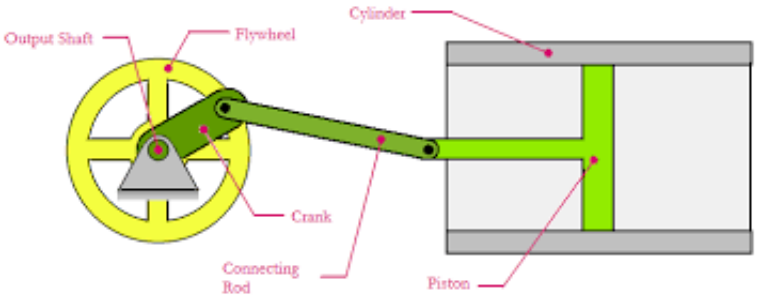

Crank and Piston

Lever: 1.
Why: the planar lever has 3DoF (two linear, one rotation). The revolute joint constrains 2DoF.

Crank and Piston: 1.
Why: Three moving links (green objects) $= 9$ DoF. The four joints restrict $4 \times 2 = 8$ DoF.

Kinematic Chains

We can combine links and joints in multiple ways:

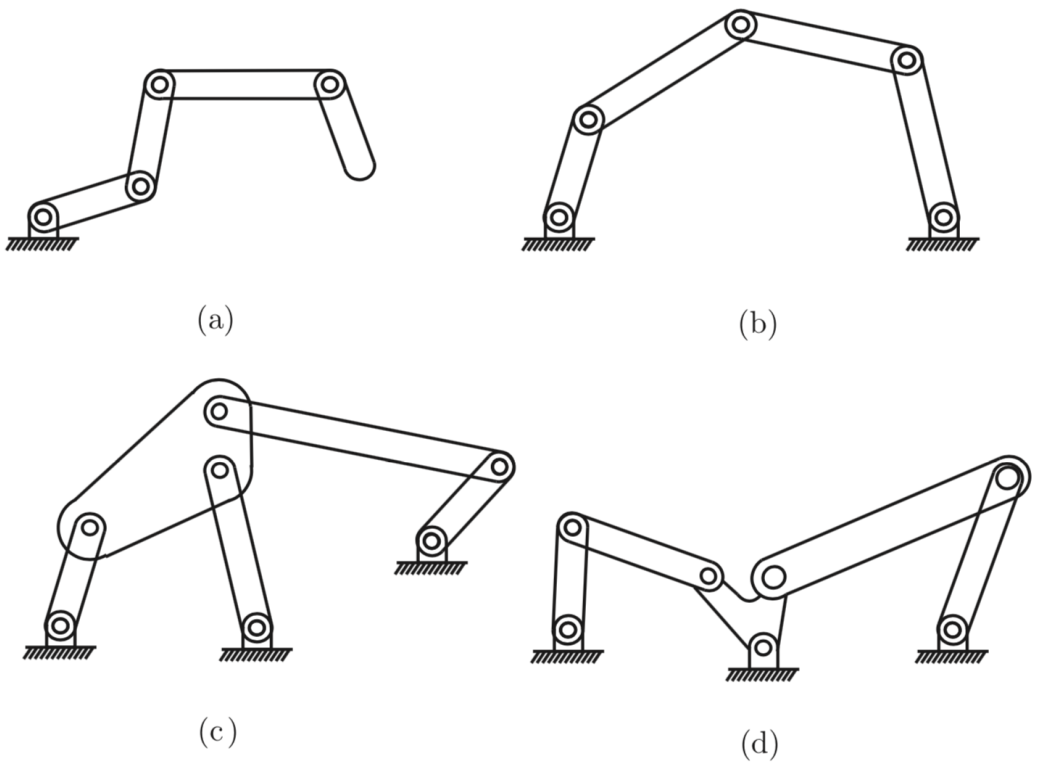

Figure 2.5: (a) k-link planar serial chain. (b) Five-bar planar linkage. (c) Stephenson six-bar linkage. (d) Watt six-bar linkage.

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Types of kinematic chains:

- Open / Closed
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Idea: think of links forming a circuit including the non-moving fixed frame. Is it open/closed? Are links in series or parallel?

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Idea: think of links forming a circuit including the non-moving fixed frame. Is it open/closed? Are links in series or parallel? Can you classify the mechanisms below?

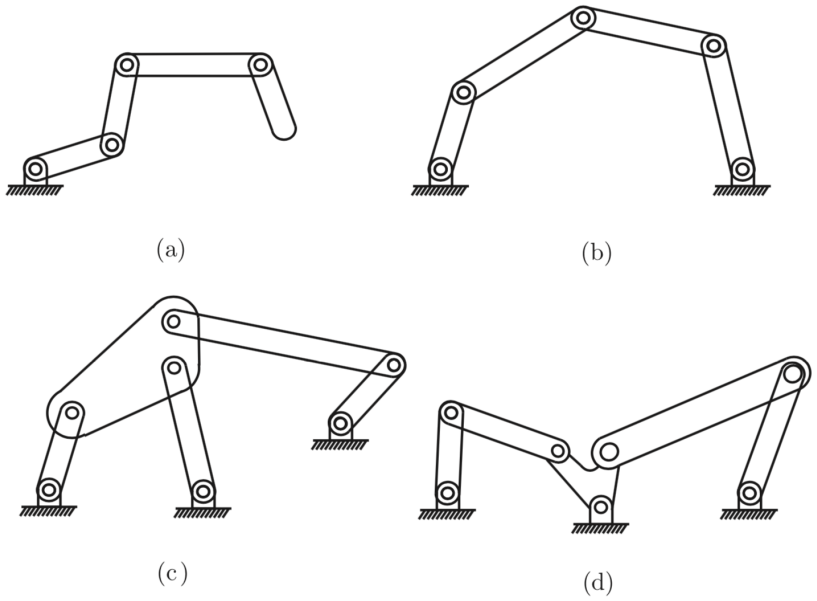

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Figure: Kinematic Chains of Rigid Bodies

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In other words, we can change and measure the configuration q of a robot with revolute or prismatic joints in practice.

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This fact is responsible for the development of robot modeling and control as it occurred, which happened before computer vision was as powerful as it is today.

Robot Configurations: Task Variables

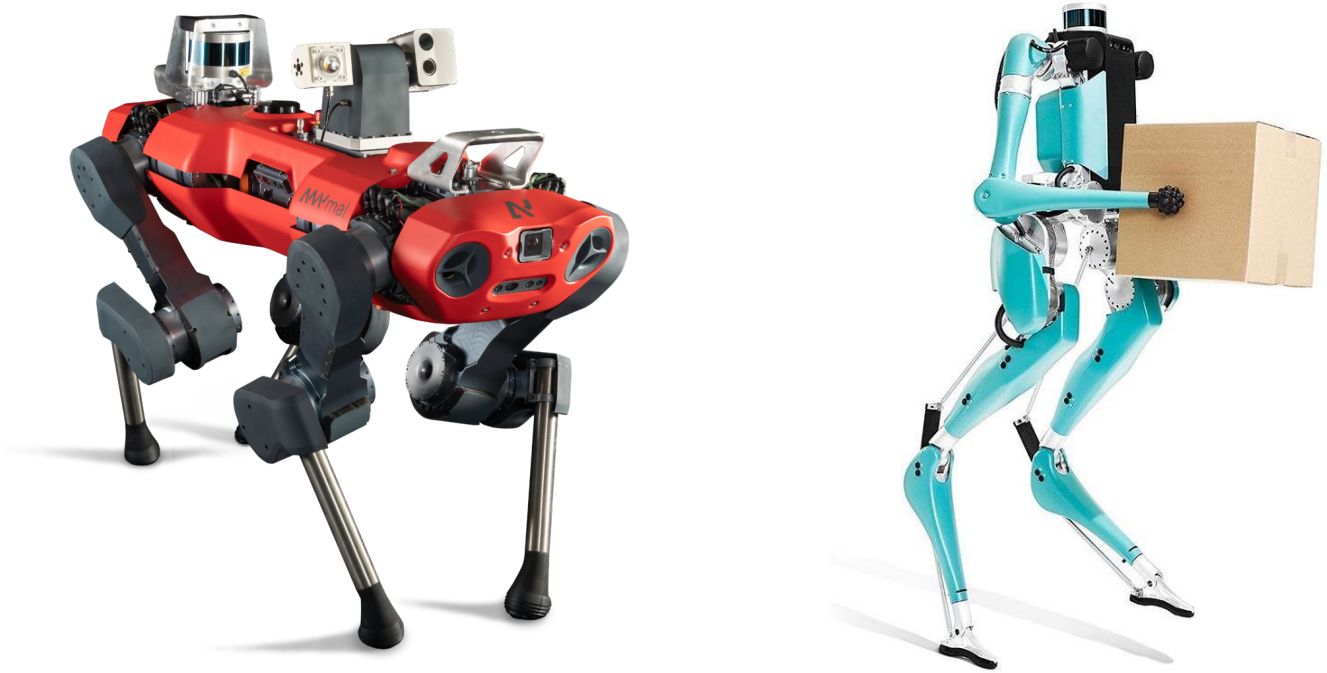

Figure: Powered Mechanisms

What do we want these powered mechanisms to do?

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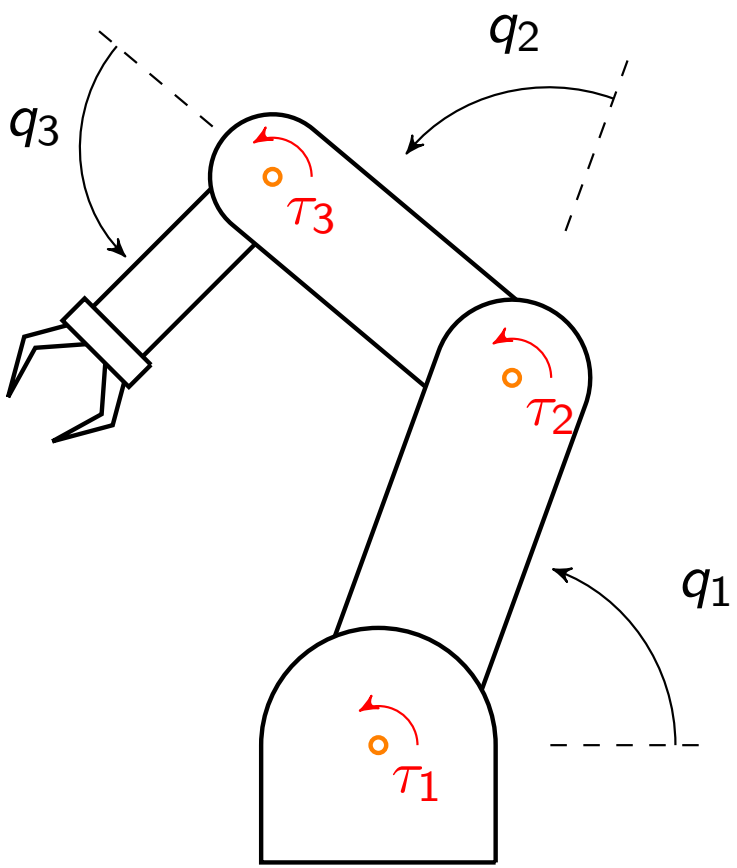
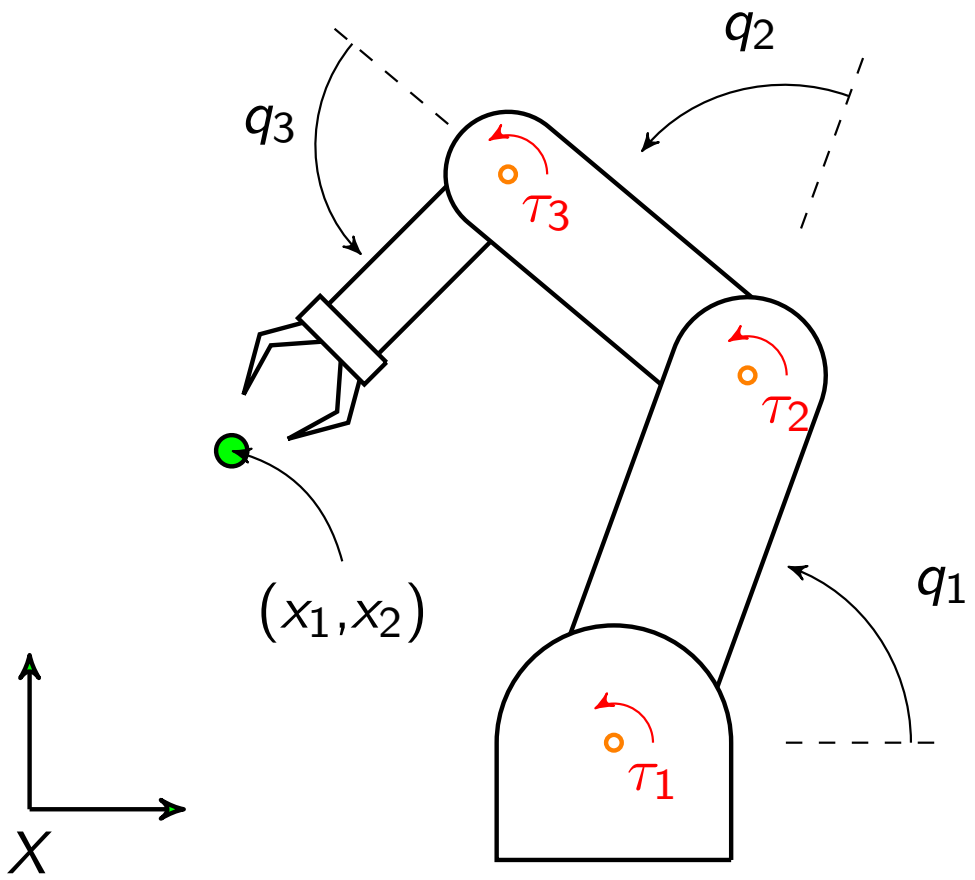


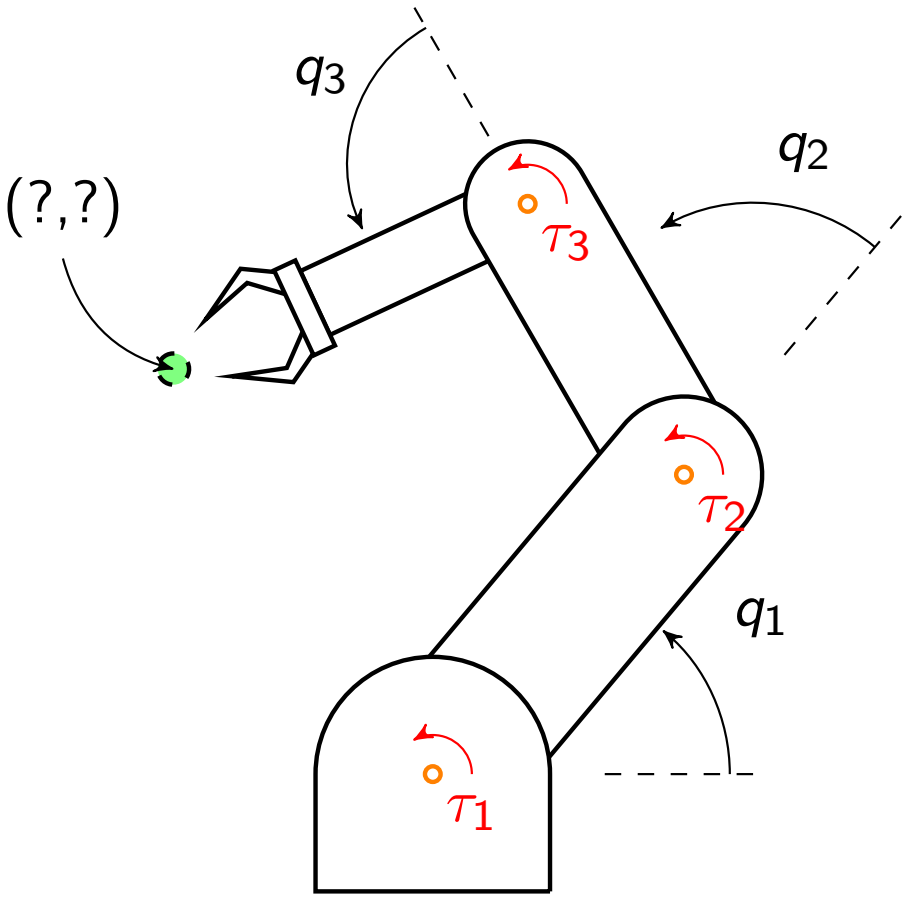
Joint Variables are 'internal' to the robot; Relevant for motion control.

Robot Configurations: Task Variables



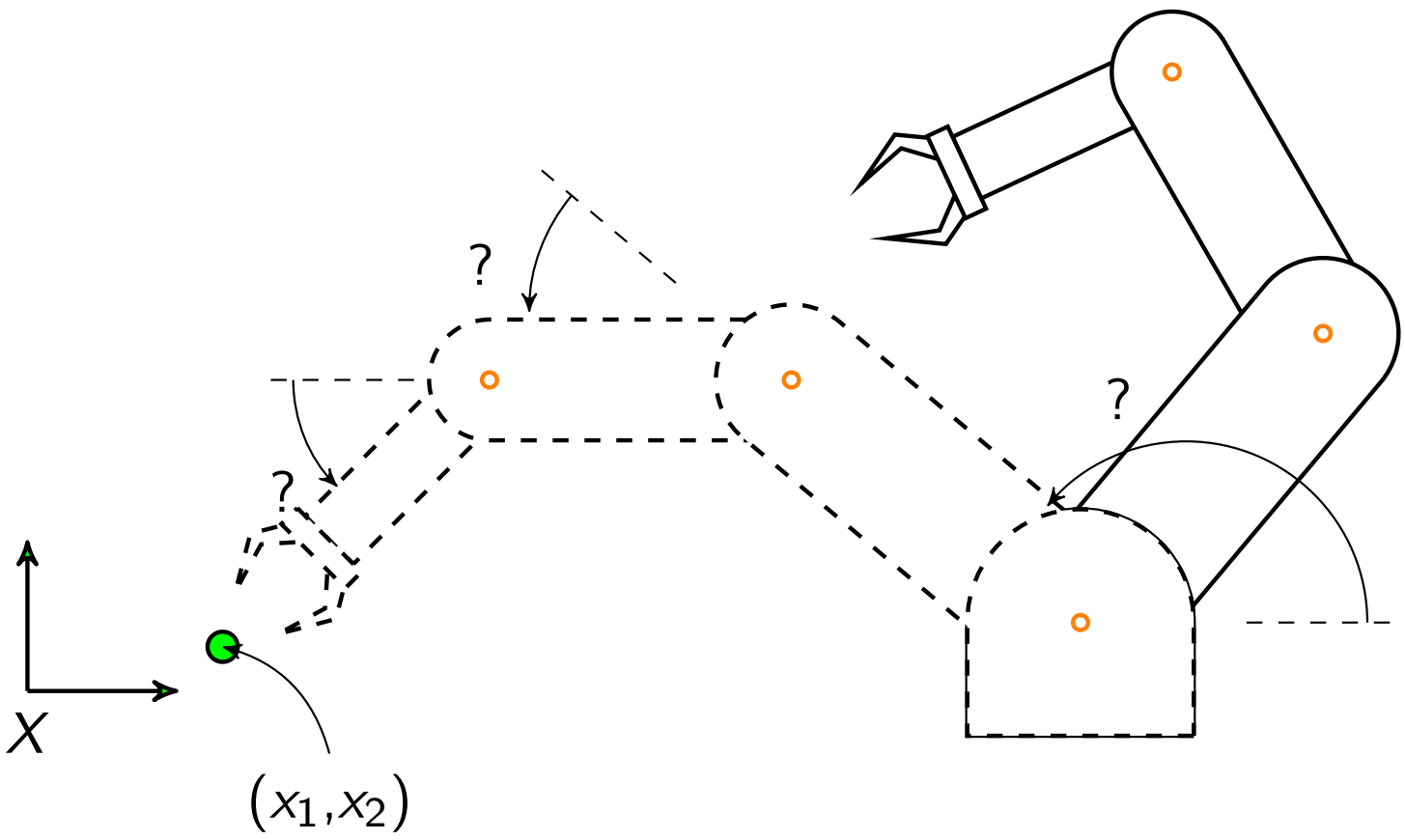
The task is to pick up an object at (x_1, x_2) defined in frame X. Therefore (x_1, x_2) forms the task coordinates χ.

Forward Kinematics



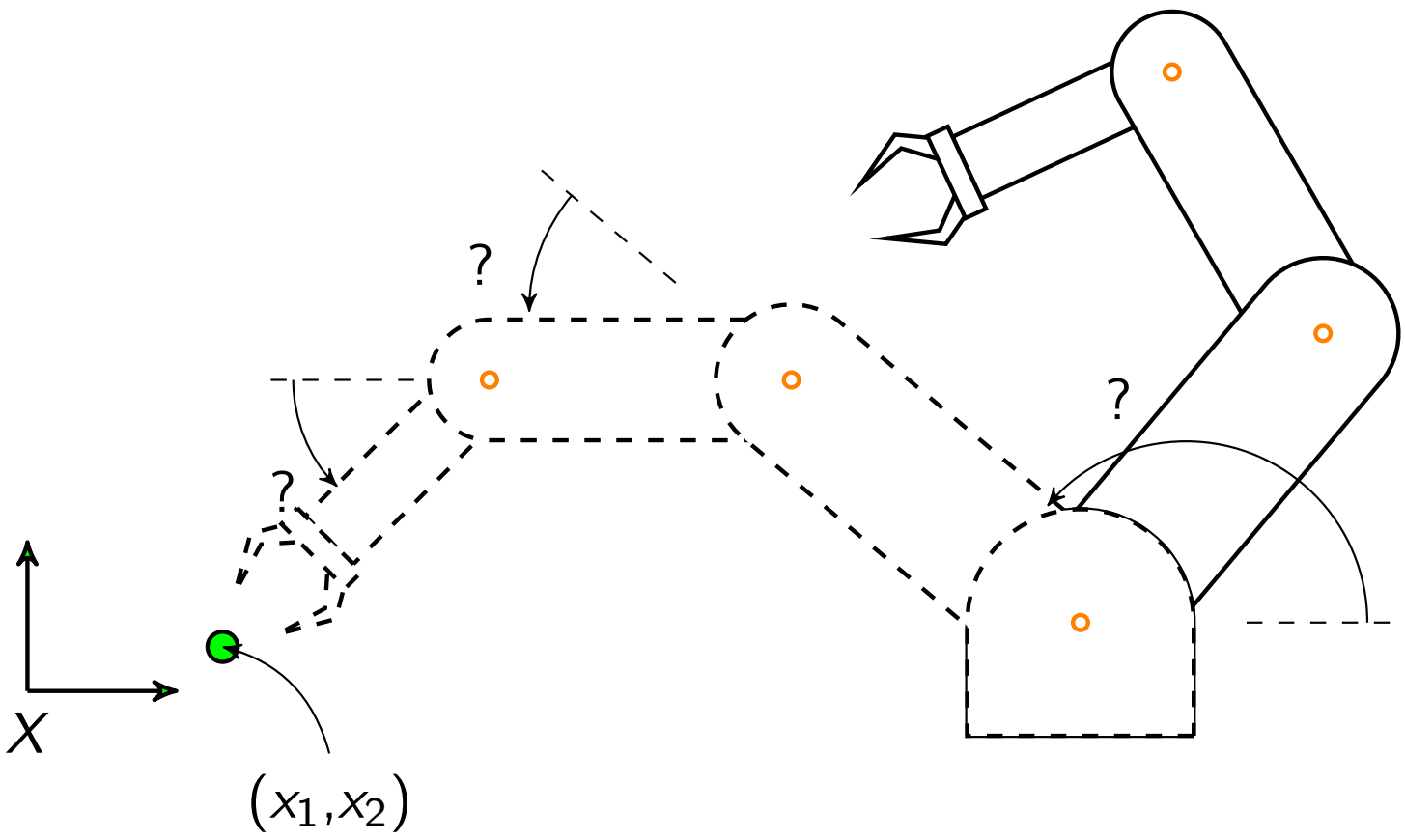
Given q, what is χ? Or, find f where $\chi = f(q)$.

Inverse Kinematics



Given χ, what is q? Or, find f^{-1} where $q = f^{-1}(\chi)$.

Inverse Kinematics



Switch between task trajectory $\chi(t)$ and joint trajectory $q(t)$.

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} = G(q) + \tau + \tau_{friction} + \tau_e, \tag{2}$$